



Assignments

Pattern Printing – 1

Q1. Print the following pattern

```
1 1 1 1
```

```
2 2 2 2
```

```
3 3 3 3
```

```
4 4 4 4
```

Solution:

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    for(int i = 1; i ≤ n; ++i) {
        for(int j = 1; j ≤ n; ++j) {
            cout << i << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Q2. Print the following pattern

Input: n = 4

Output:

```
1 2 3 4
```

```
1 2 3
```

```
1 2
```

```
1
```

Solution:

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    for(int i = 1; i ≤ n; ++i) {
        for(int j = 1; j ≤ n-i+1; ++j) {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Q3. Print the following pattern

Input: n = 4

Output:

A

A B

A B C

A B C D

Solution:

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    for(int i = 1; i ≤ n; ++i) {
        for(int j = 1; j ≤ i; ++j) {
            cout << (char)('A'+j-1) << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Q4. Print the following pattern

Input: n = 4

Output:

```
1
A B
1 2 3
A B C D
1 2 3 4 5
```

Solution:

```
#include <bits/stdc++.h>
int main() {
    int n;
    cin >> n;
    for(int i = 1; i ≤ n; ++i) {
        if(i % 2 == 1)
            for(int j = 1; j ≤ i; ++j) {
                cout << j << " ";
            }
        else
            for(int j = 1; j ≤ i; ++j) {
                cout << (char) ('A' + j - 1) << " ";
            }
        cout << endl;
    }
    return 0;
}
```

Q5. Print the following pattern

Input: n = 4

Output:

```
*  
  
**  
  
***  
  
****  
  
***  
  
**  
  
*
```

Solution:

```
#include <bits/stdc++.h>  
int main() {  
    int n;  
    cin >> n;  
    // Printing upper triangle  
    for(int i = 0; i < n; ++i) {  
        for(int j = 0; j ≤ i; ++j) {  
            cout << "* ";  
        }  
        cout << endl;  
    }  
    // Printing lower triangle  
    for(int i = 0; i < n-1; ++i) {  
        for(int j = 0; j < n - i - 1; ++j) {  
            cout << "* ";  
        }  
        cout << endl;  
    }  
}
```

Q6. Print the following pattern

Input : m = 4, n = 6

Output :

```
*****  
  
*      *  
  
*      *  
  
*****
```

Solution:

```
#include <iostream>  
using namespace std;  
int main() {
```

```
int m, n;
cin >> m >> n;
for(int i = 0; i < m; ++i) {
for(int j = 0; j < n; ++j) {
if(i == 0 || j == 0 || i == m-1 || j == n-1) {
cout << "* ";
} else {
cout << " ";
}
}
cout << endl;
}
return 0;
}
```

Q7. Print the following pattern

Input : m = 4, n = 6

Output :

```
****
****
****
****
```

Solution:

```
#include <iostream>
using namespace std;
int main() {
int n;
cin >> n;
for(int i = 0; i < n; ++i) {
for(int j = 0; j < n-i-1; ++j) {
cout << " ";
}
for(int j = 0; j < n; ++j) {
cout << "* ";
}
cout << endl;
}
return 0;
}
```

Q8. Print the following pattern

Input : n= 4

Output :

1

12

123

1234

Solution:

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    for(int i = 0; i < n; ++i) {
        for(int j = 0; j < n-i-1; ++j) {
            cout << " ";
        }
        for(int j = 1; j ≤ i+1; ++j) {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Q9. Print the following pattern

Input : m = 4, n = 6

Output :

A

A B

A B C

A B C D

Solution:

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    for(int i = 0; i < n; ++i) {
        for(int j = 0; j < n-i-1; ++j) {
            cout << " ";
        }
    }
```

```
for(int j = 1; j ≤ i+1; ++j) {  
    cout << (char)('A' + j - 1) << " ";  
}  
cout << endl;  
}  
return 0;  
}
```

Q10. Print the following pattern

Input : m = 4, n = 6

Output :

1

2 1

3 2 1

4 3 2 1

Solution:

```
#include <iostream>  
using namespace std;  
int main() {  
    int n;  
    cin >> n;  
    for(int i = 0; i < n; ++i) {  
        for(int j = 0; j < n-i-1; ++j) {  
            cout << " ";  
        }  
        for(int j = i+1; j ≥ 1; --j) {  
            cout << j << " ";  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

Q10. Print the following pattern

Input : m = 4, n = 6

Output :

*

**

**

*

Solution:

```
#include <bits/stdc++.h>
int main() {
    int n;
    cin >> n;
    // Printing upper triangle
    for(int i = 0; i < n; ++i) {
        for(int j = 0; j < n-i-1; ++j) {
            cout << " ";
        }
        for(int j = 0; j ≤ i; ++j) {
            cout << "* ";
        }
        cout << endl;
    }
    // Printing lower triangle
    for(int i = 0; i < n-1; ++i) {
        for(int j = 0; j ≤ i; ++j) {
            cout << " ";
        }
    }
```

Note:- Please try to invest time doing the assignments which are necessary to build a strong foundation. Do not directly Copy Paste using Google or ChatGPT. Please use your brain.